

$5.67 \rightarrow 6 > 5 ; 5.67 \rightarrow 6$
 $5.67 \rightarrow 7 > 5 \checkmark$
 $5.7 \rightarrow 6$

(OFF) $\rightarrow 0.C$
 FALSE $\leftarrow 0 \rightarrow$ Binary zero and $1 \rightarrow$ Binary one $\rightarrow$ TRUE.

Continuous time domain system

(ON) $\rightarrow 5.C$

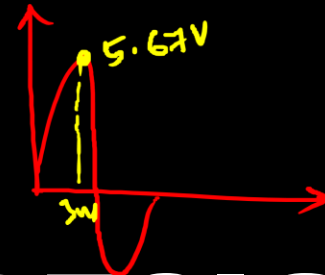
DIGITAL

PUN
PDN
(1)

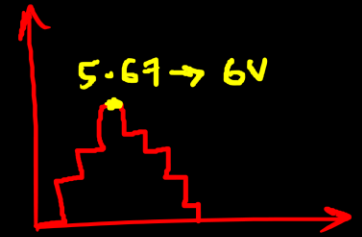


Pseudo NMOS / PMOS (3)

domino dynamic CMOS (4)



"ADC" / Encoding - decoding + Quantizing
 (10) (5) (5) $\rightarrow 1$

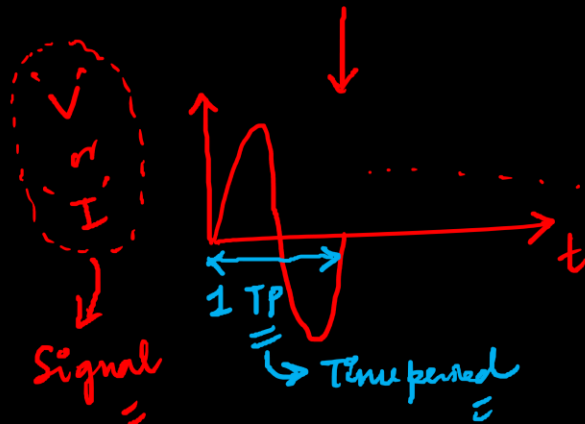


ANALOG VLSI DESIGN & CHAR.

Er. Sailesh Mishra

B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI



[Mathematical Function]

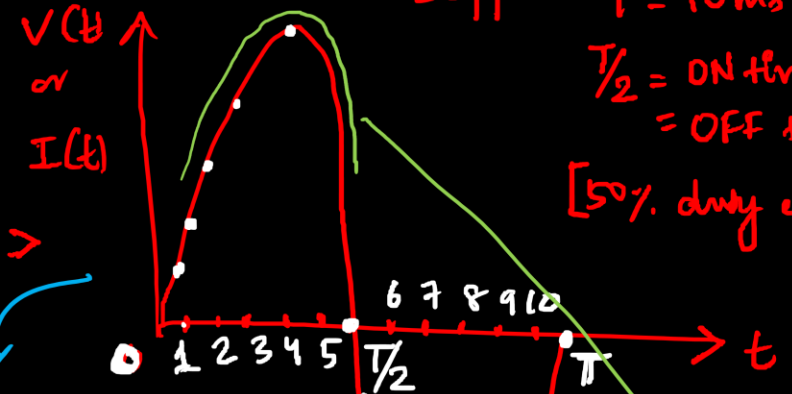
eg $\rightarrow 2\pi f_c \cdot t$

$\sin(2\pi f_c \cdot t), \cos(2\pi f_c \cdot t)$

HARMONIC FUNCTION $\rightarrow$ SHM $\rightarrow$ PERIODIC (repeats at Regular interval of time).

$f_c = \frac{1}{T} \rightarrow$ time period

$V(t)$
or
 $I(t)$



ANALOG Signal

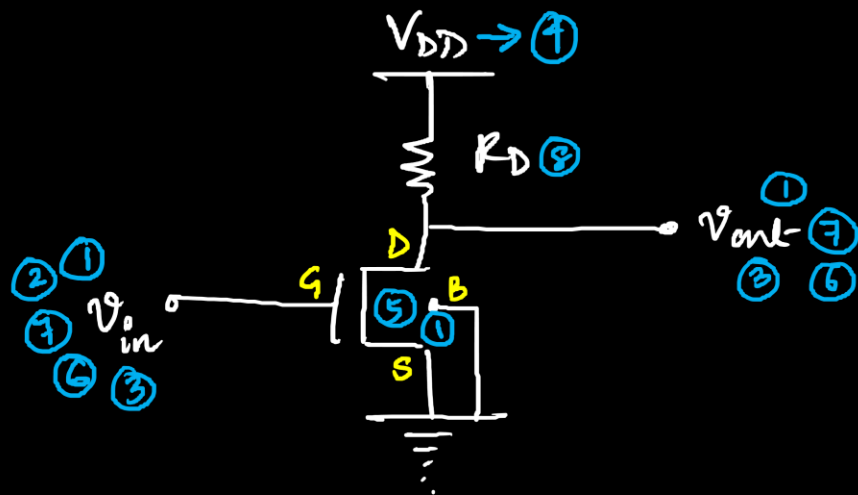
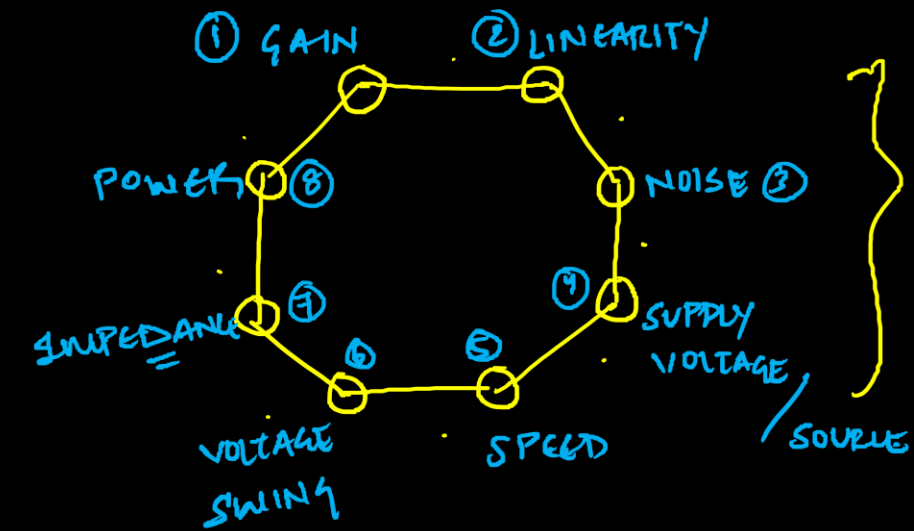
Envelope of a Signal

Suppose $T = 10ms$

$T/2 =$ ON time
 $=$ OFF time

[50% duty cycle]

ANALOG DESIGN OCTAGON :-



→ RESISTIVE LOAD ←
Common Source Stage Amplifier

→ To Implement & Execute a MOSFET as an Amplifier;
We Need to BIAS it / OPERATE it in "Saturation"!!

→ MOSFET $\begin{cases} \text{PMOS} \\ \text{NMOS} \end{cases} \rightarrow \text{ES's} \rightarrow \text{PMOS.}$

→ WHY SATURATION?
↓

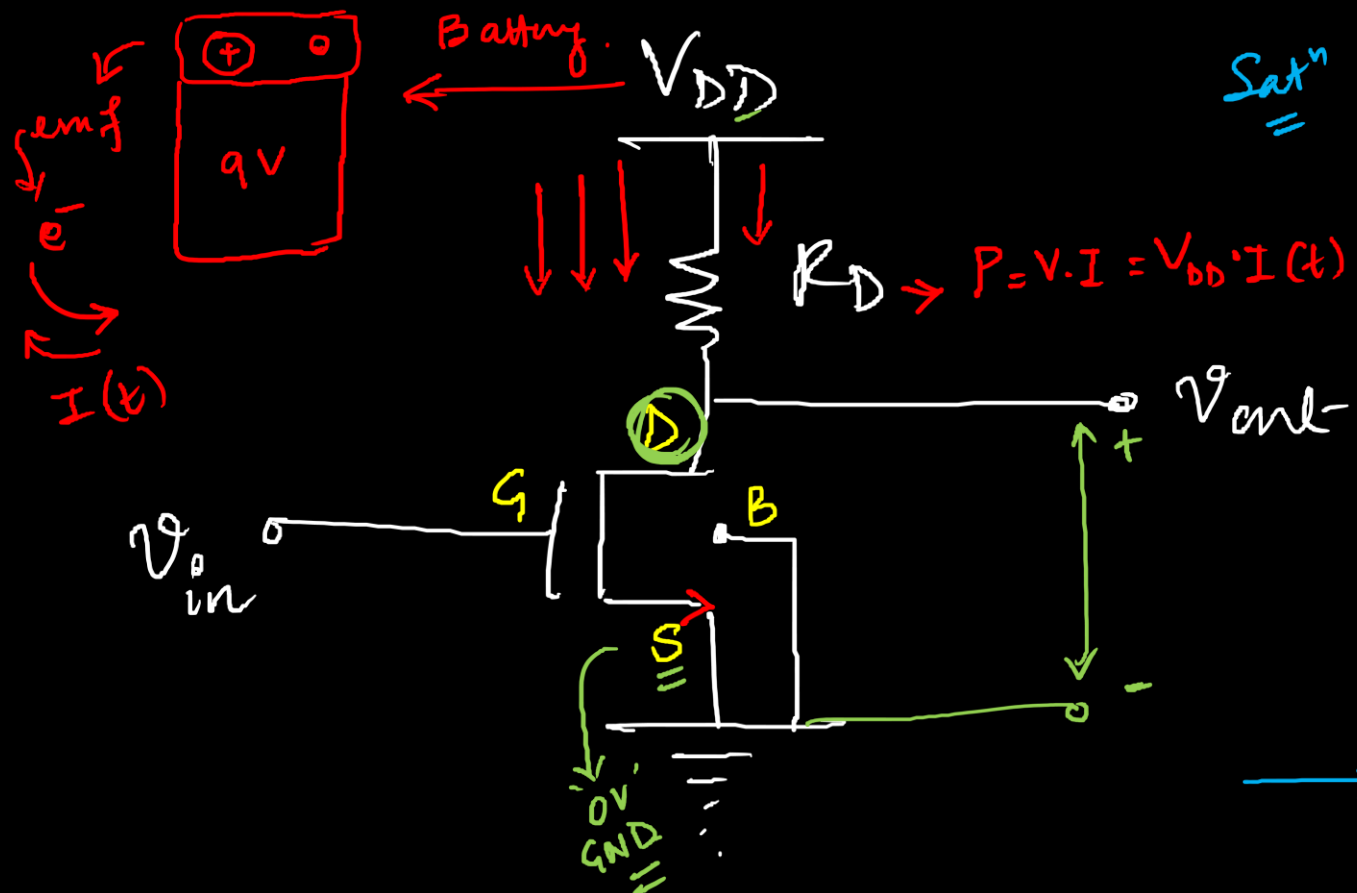
Because in "Sat"; MOSFET
have Better Current driving
Capability.

$$V_{SD} > V_{SG} + |V_{tp}|$$

NMOS.

$$V_{DS} > V_{GS} - V_{tn}$$

⊙ MATHEMATICAL ANALYSIS (P.T.O)



Satⁿ $\div$

$$[V_{DS} > V_{GS} - V_{th} - I_D.R_D]$$

o/p
Terminal
voltage
wrt
GND
=
or
SOURCE
=

is
Basically
the
i/p
signal
or
voltage.
=
i.e (v_{in})
=

Threshold
voltage of
the device
=
always
[constant]
=

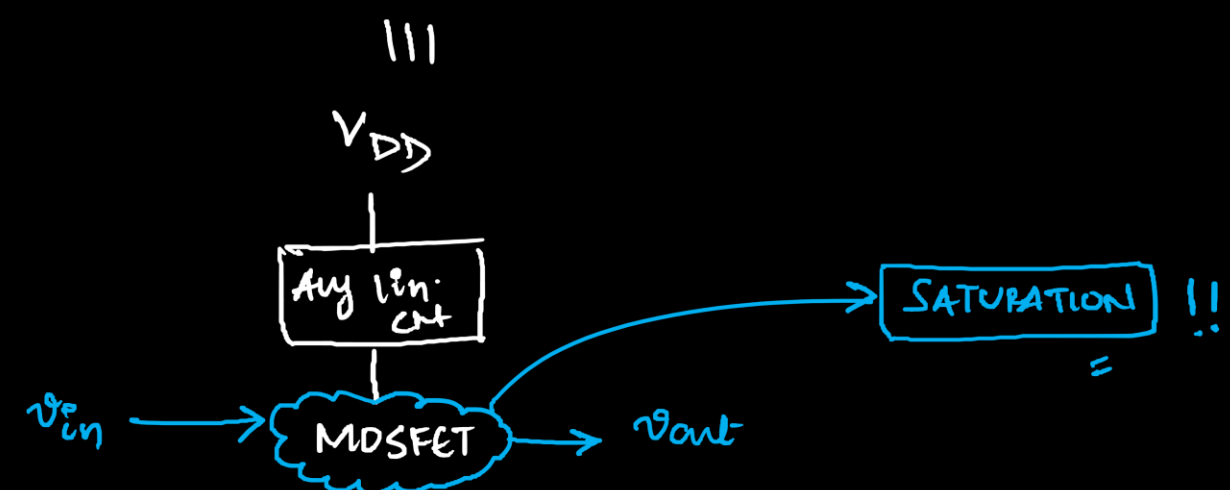


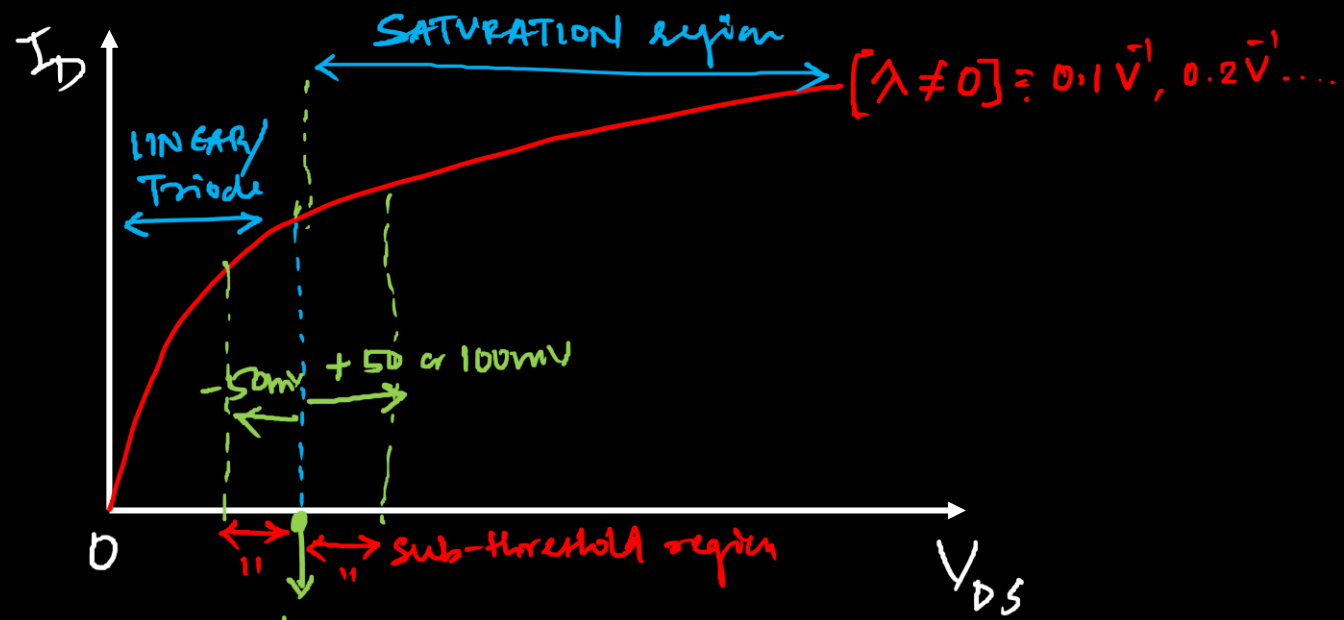
$$V_{DS} > V_{GS} - V_{th} - I_D.R_D$$

"Equal to" physically means, I am
operating MOSFET @ edge of linear

Satⁿ
=

low
power VLSI design
=





$V_{GS} - V_{th}$
 (MARGIN of
 linear region &
 Satⁿ region).

► Example 2.2

On a V_{DS} - V_{GS} plane, show the regions of operation of an NMOS transistor.

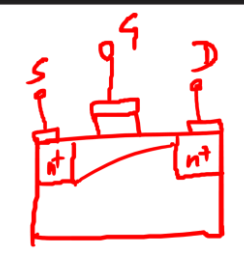
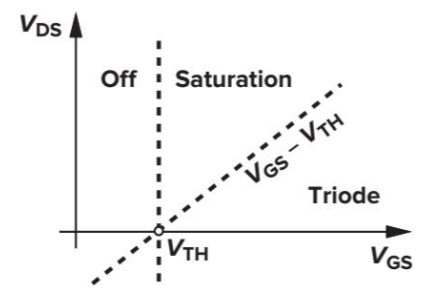
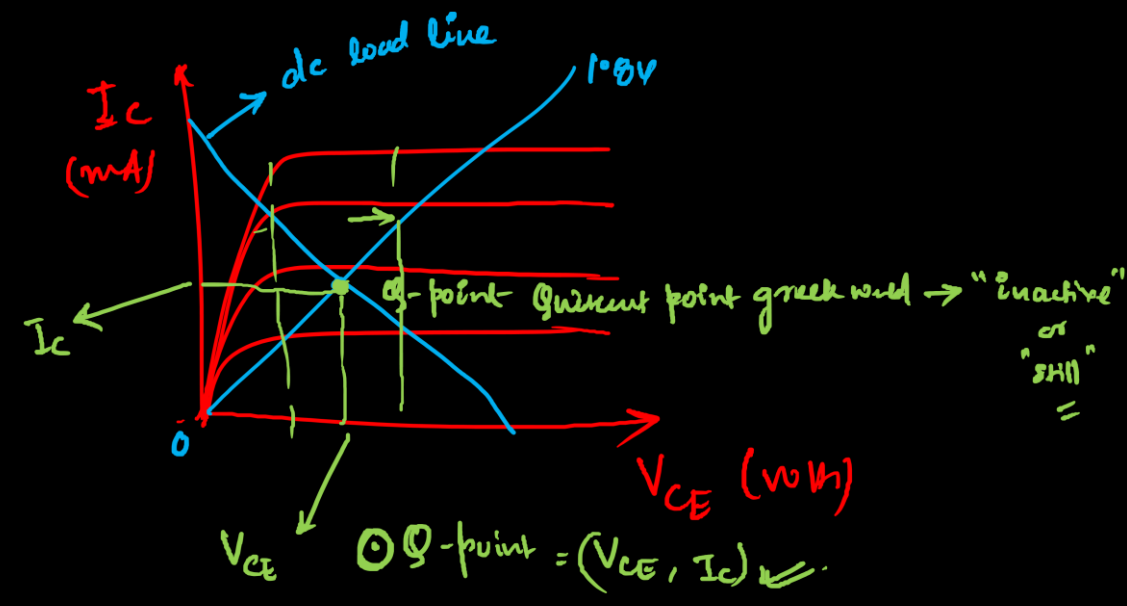


Figure 2.18 V_{DS} - V_{GS} plane showing regions of operation.

Solution

Since the value of V_{DS} with respect to $V_{GS} - V_{TH}$ determines the region of operation, we draw the line $V_{DS} = V_{GS} - V_{TH}$ in the plane, as shown in Fig. 2.18. If $V_{GS} > V_{TH}$, then the region above the line corresponds to saturation, and that below the line corresponds to the triode region. Note that for a given V_{DS} , the device eventually leaves saturation as V_{GS} increases. The minimum allowable V_{DS} for operation in saturation is also called $V_{D,sat}$. It is important to bear in mind that $V_{D,sat} = V_{GS} - V_{TH}$.



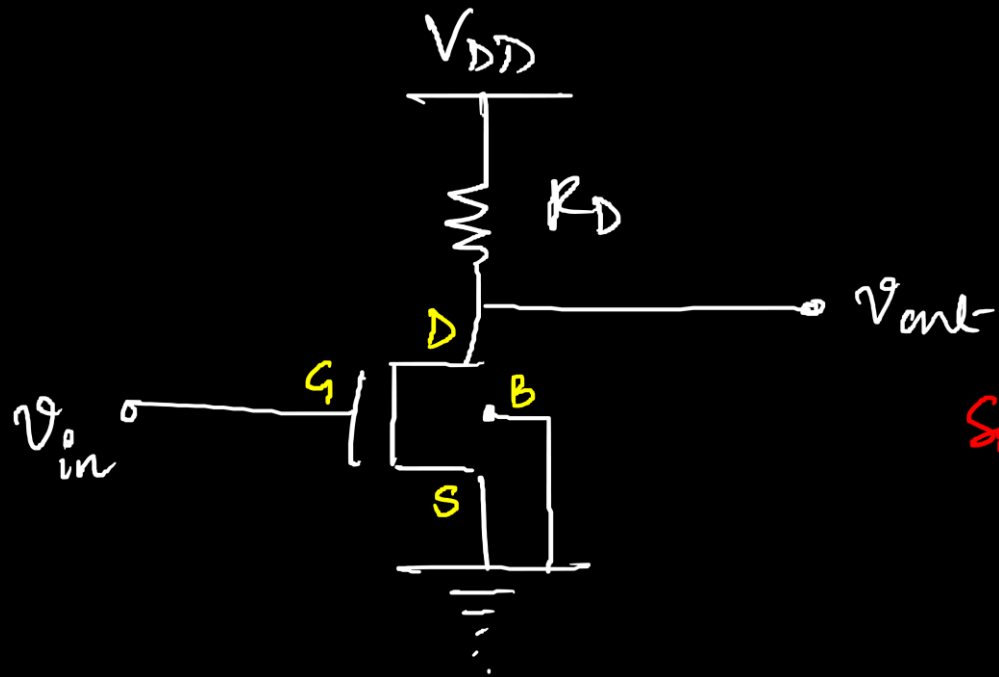
HOW TO OPERATE MOSFET IN SATURATION?

INPUT / Calibration =
GATE
VOLTAGE

$$\rightarrow V_{th} = 600 \text{ mV} = 0.6 \text{ V} \rightarrow \text{CONSTANT !!}$$

$$\rightarrow V_{GS} = V_{th} + 200 \text{ mV} = 600 \text{ mV} + 200 \text{ mV} = 800 \text{ mV} = \boxed{0.8 \text{ V}}$$

$$\rightarrow V_{DS} = V_{GS} + 210 \text{ mV} = 800 \text{ mV} + 210 \text{ mV} = 1010 \text{ mV} = \boxed{1.01 \text{ V}}$$



$$\underline{\text{SAT}}^n \rightarrow V_{DS} > V_{GS} - V_{th}$$

$$= 1.01 \text{ V} > 0.8 - 0.6 \text{ V}$$

$$= 1.01 \text{ V} > 0.2 \text{ V} \checkmark \rightarrow \text{SATURATION.}$$

So, From the Above Analysis of Bias point Calibration,

V_{DS} is Much More; Hence; Power Consumption will be proportionally More !! ; So, We'll decrease it by

Keeping a Margin of $\approx 50-100 \text{ mV}$ or 200 mV .

$$(V_{DS})_{\text{New}} = (V_{GS} - V_{th}) + 200 \text{ mV} = 0.2 + 200 \text{ mV} = \boxed{0.4 \text{ V}} \quad !!$$

Now; Finally we've calibrated the device with the following Terminal Potentials as Below:-

(i) GATE VOLTAGE $\rightarrow V_{in} \rightarrow V_{GS} = \boxed{0.8V}$ ✓

(ii) DRAIN VOLTAGE $\rightarrow V_{out} \rightarrow V_{DS} = \boxed{0.4V}$ ✓

$$(I_D)_{sat} = \frac{1}{2} \mu_n C_{ox} \cdot \frac{W}{L} \{V_{GS} - V_{th}\}^2$$

$$= \frac{1}{2} \times \frac{200 \mu A}{V^2} \times \frac{10}{1} \{0.8V - 0.6V\}^2$$

$$= 0.5 \times \frac{200 \mu A}{V^2} \times 10 \times (0.2)^2$$

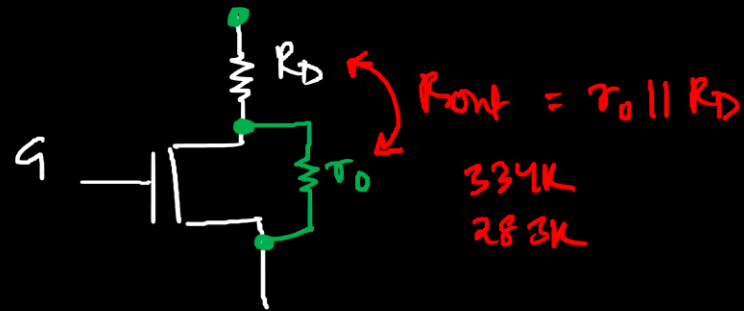
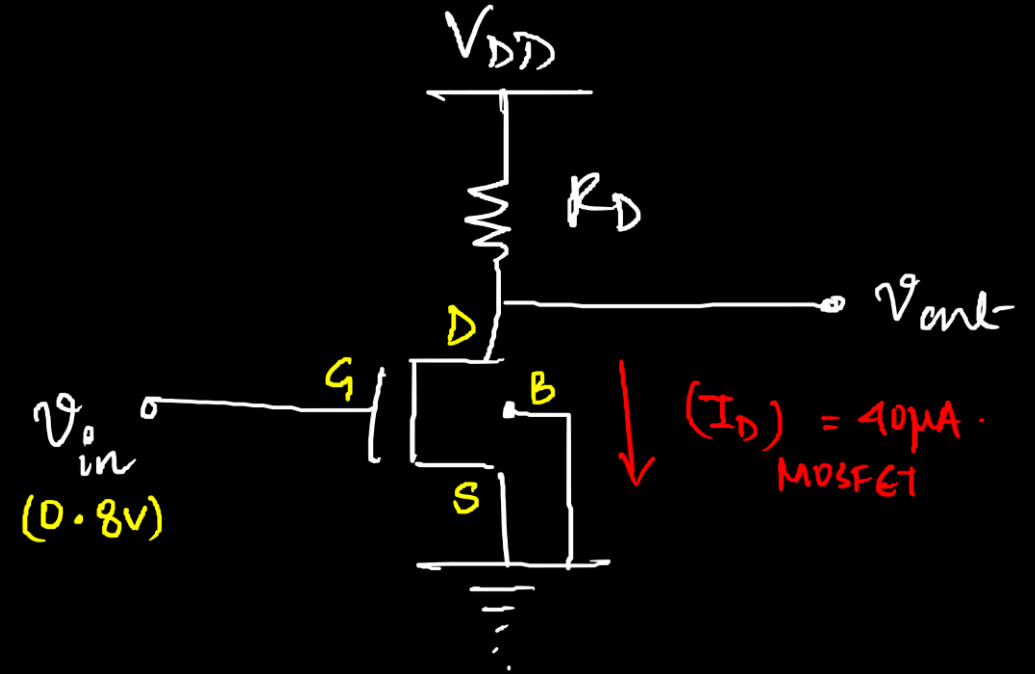
$$\Rightarrow \boxed{(I_D)_{sat} = 40 \mu A} =$$

We know;

$$A_v = g_m \cdot R_D \text{ or } [g_m] R_{out}$$

$\rightarrow$ MOSFET'S
transconductance.

① $\underline{g_m}$ $\rightarrow$ total ckt's
TRANSCONDUCTANCE.

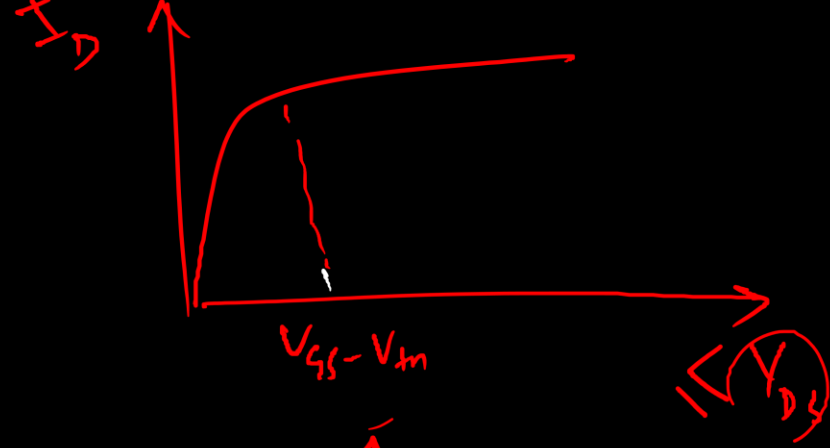


What is TRANS CONDUCTANCE ?

TRANS
(transfer)
=

CONDUCTANCE
(Conductivity).
=

the extent/ability to conduct flawlessly.
=



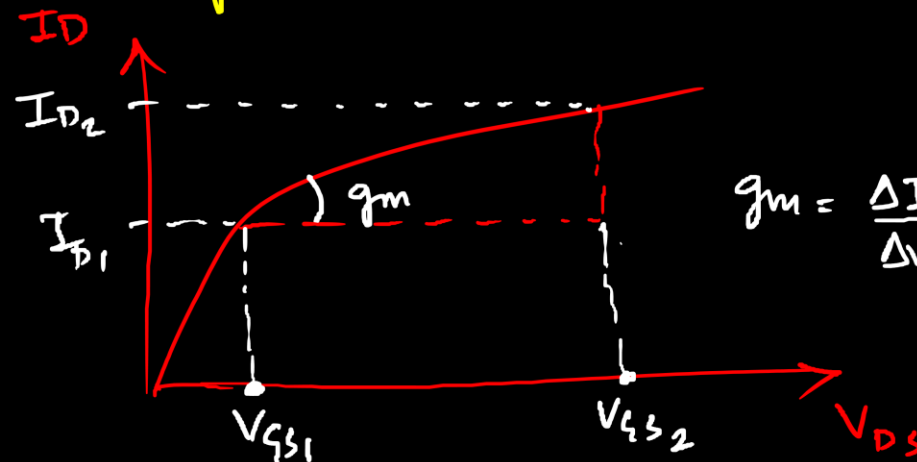
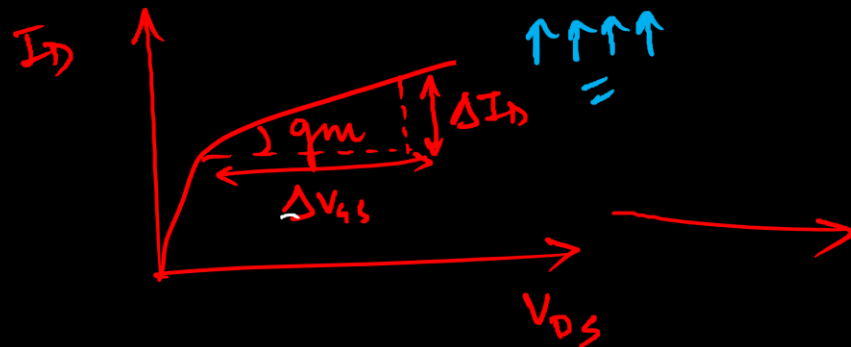
①
(current dividing
capability)

$$\Rightarrow A_v = g_m \cdot R_{out}$$

① $g_m = \frac{\Delta I_D}{\Delta V_{GS}}$ ✓

o/p voltage
↑↑↑↑

② $g_m = \frac{\Delta I_D}{\Delta V_{DS}}$
or
 g_{DS}

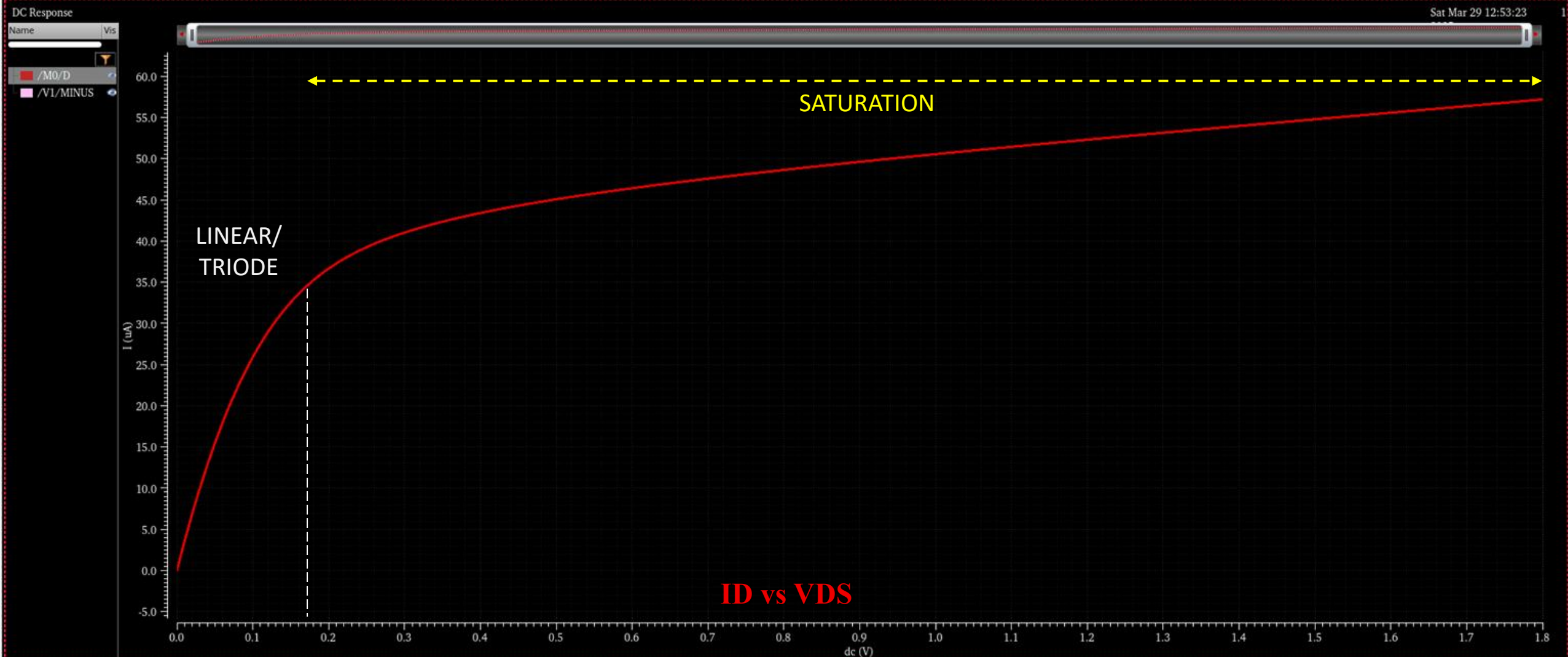


$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{I_{D2} - I_{D1}}{V_{GS2} - V_{GS1}} = \frac{10 \text{ mA} - 8 \text{ mA}}{V_{GS2} - V_{GS1}}$$





sailesh_nmos_char_adi_1 ×

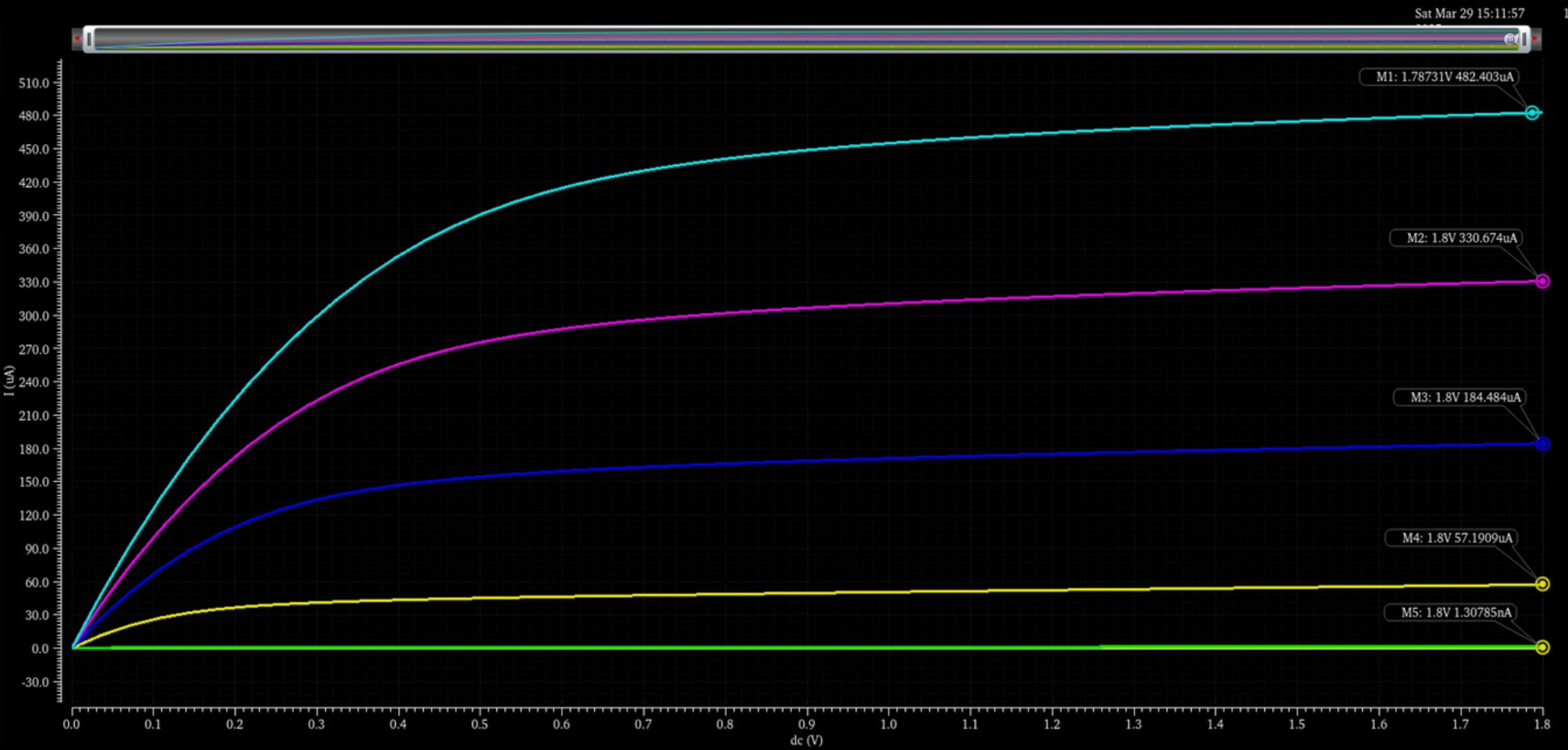


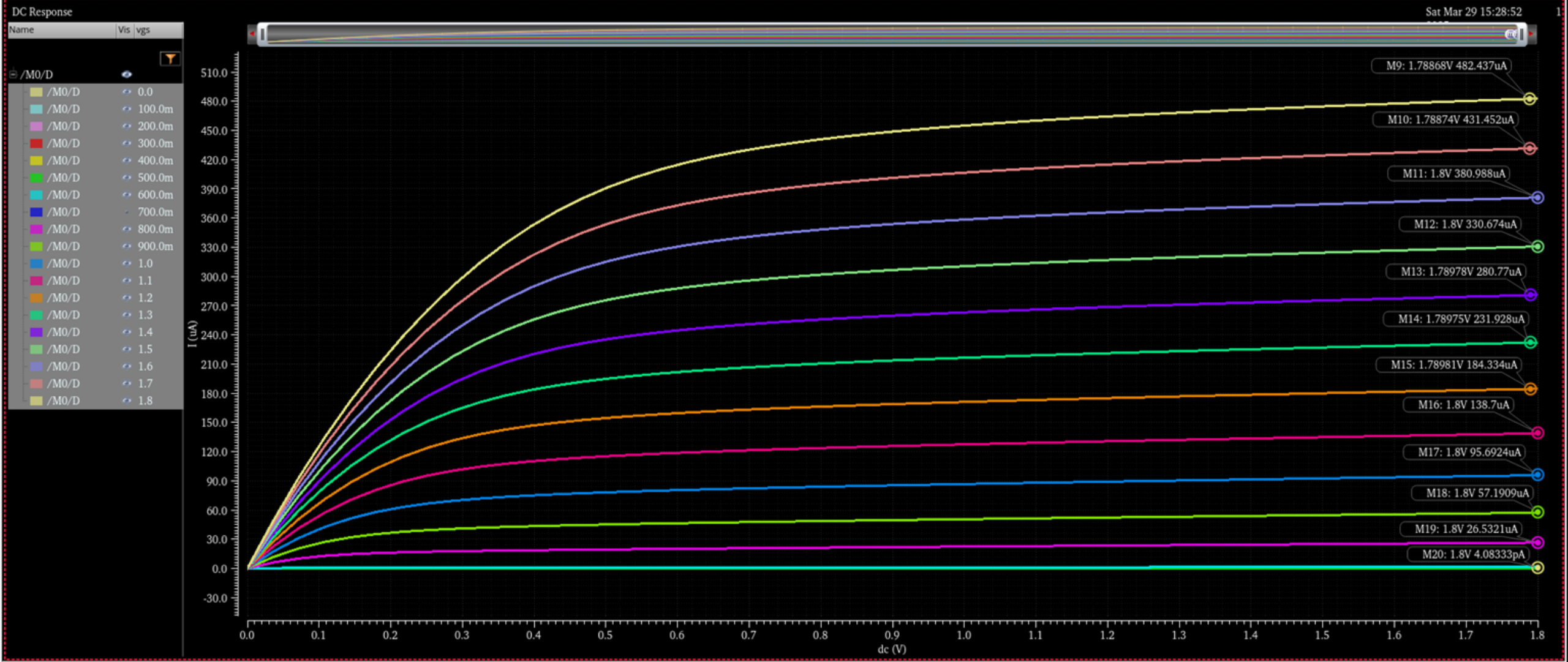
DC Response

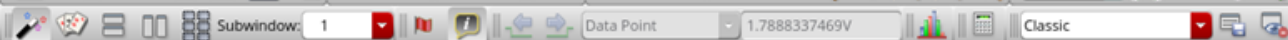
Name Vis vgs

/M0/D

- /M0/D 0.0
- /M0/D 300.0m
- /M0/D 600.0m
- /M0/D 900.0m
- /M0/D 1.2
- /M0/D 1.5
- /M0/D 1.8







saillesh nmos_char_adi schematic ×

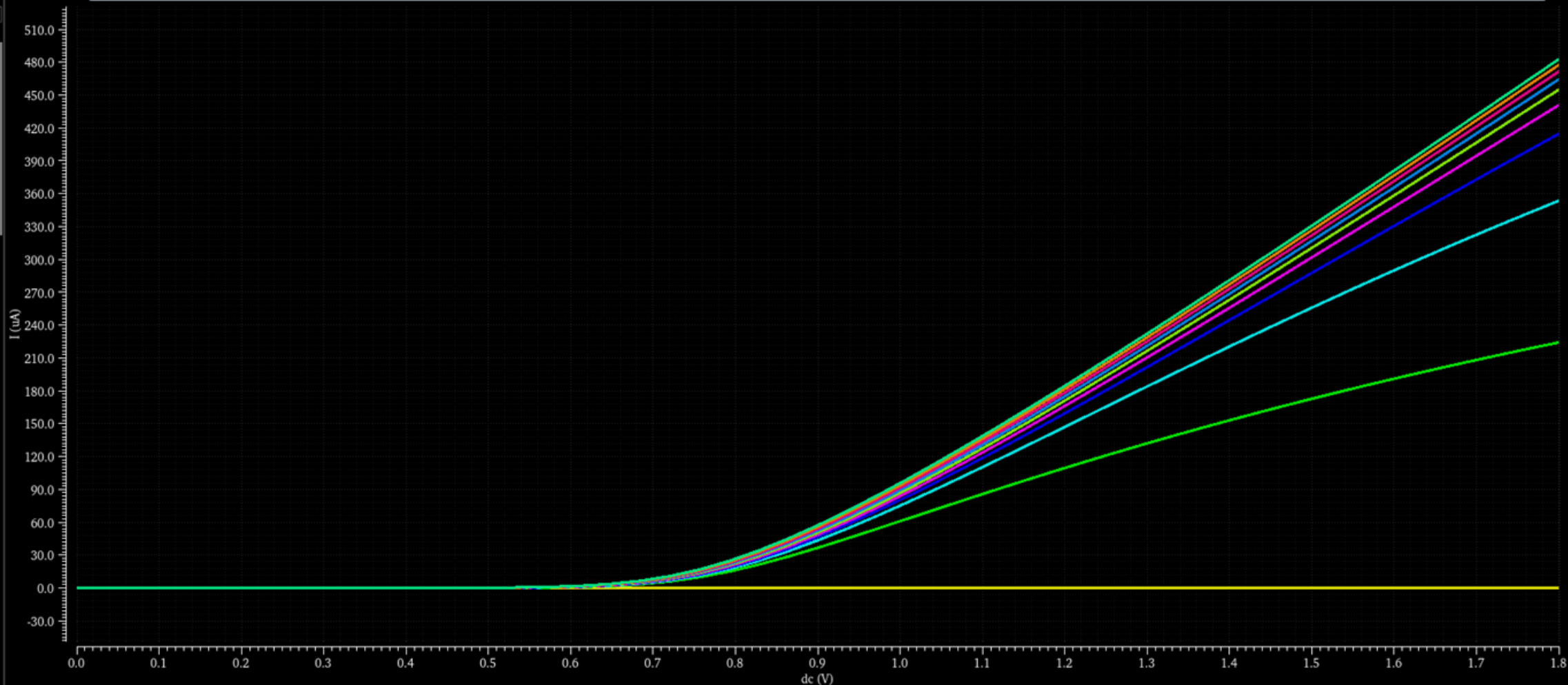
DC Response

Name Vis vds

/M0/D

/M0/D	0.0
/M0/D	200.0m
/M0/D	400.0m
/M0/D	600.0m
/M0/D	800.0m
/M0/D	1.0
/M0/D	1.2
/M0/D	1.4
/M0/D	1.6
/M0/D	1.8

Sat Mar 29 16:30:23



mouse L:

M:

R:

32(78) Trace: /M0/D (vds=1.8); Context: /home/local/simulation/Vlab14/nmos_char_adi/spectre/schematic/psf; Dataset: dc-dc

Virtuoso (R) Visualization & Analysis XL



cadence

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help



sailesh nmos_char_adi schematic

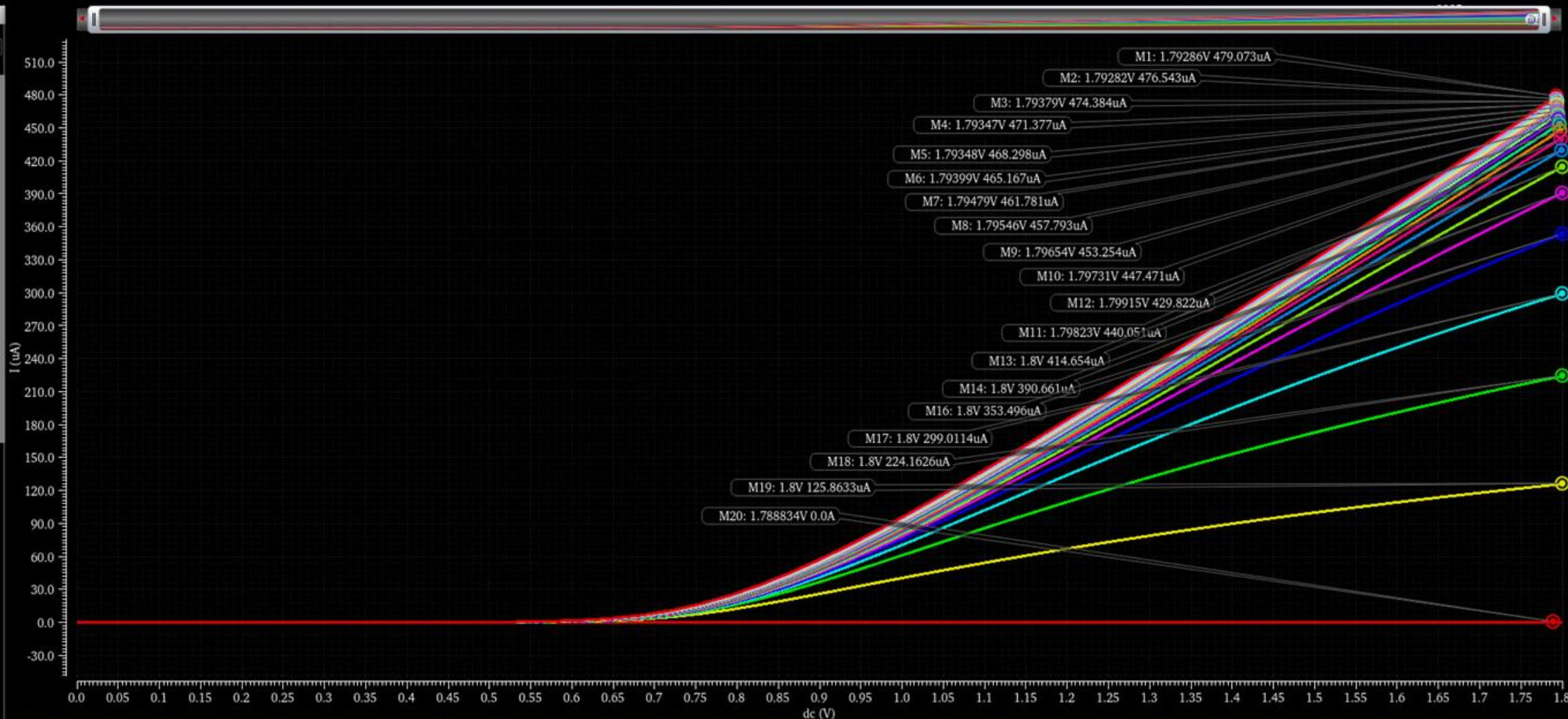
DC Response

Name Vis vds

/M0/D

- /M0/D 0.0
- /M0/D 100.0m
- /M0/D 200.0m
- /M0/D 300.0m
- /M0/D 400.0m
- /M0/D 500.0m
- /M0/D 600.0m
- /M0/D 700.0m
- /M0/D 800.0m
- /M0/D 900.0m
- /M0/D 1.0
- /M0/D 1.1
- /M0/D 1.2
- /M0/D 1.3
- /M0/D 1.4
- /M0/D 1.5
- /M0/D 1.6
- /M0/D 1.7
- /M0/D 1.8

Sat Mar 29 16:22:40



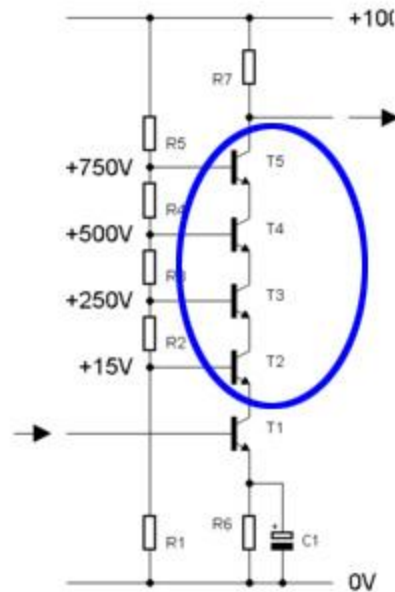
mouse L:

M:

R:

32(78) Trace: /M0/D (vds=0); Context: /home/local/simulation/vlab14/nmos_char_adi/spectre/schematic/psf; Dataset: dc-dc

Cascoding Amp.
Transistor Loads are stacked
Vertically.



Cascading Amp.
Transistor Loads are
daisy chained horizontally.

